

Quiz 1 Review

Name: _____

1. Please complete the following from memory

A. $1 \text{ AU} = \text{_____ m.}$

D. $R_{\odot} = \text{_____ kg.}$

B. $1 \text{ ly} = \text{_____ m.}$

E. Area of a Circle, $A = \text{_____}.$

C. $M_{\oplus} = \text{_____ m.}$

F. $M_{\text{yr}} = \text{_____ yr}$

2. The Moon appears so much larger than planet Jupiter in the sky because it _____.

A. lies so much closer to Earth

C. lies so much further away from Earth

B. is so much bigger than Jupiter

D. is so much smaller than Jupiter

3. The star Betelgeuse is about 642 light years from the Earth. How far is this in meters?

4. Simplify the following:

A. $10^3 \times 10^4 \times 10^5 =$

B. $10^6 \div 10^{-1} =$

C. $(10^3)^{-2} =$

D. Solve for x: $x^{3.6} = y^7 =$

5. Pharmacologists use the following equation to determine the proper dosage, D, for the drugs they administer, given a patient's weight, W, and given a clearance rate, C (the rate at which the body is able to metabolize the drug).

$$D^2 = 1.5 W^{1.5} C$$

A. Write a proportionality relation between D and W

B. If the weight of the patient is doubled, and the clearance rate is tripled, the dosage would have to increase by a factor of _____ times.